

Rayleigh–Bénard Convection

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Foundation models for spatial-time series
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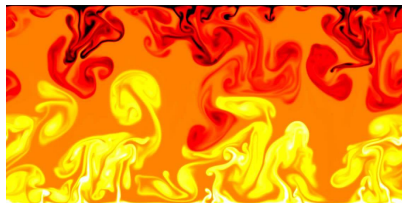
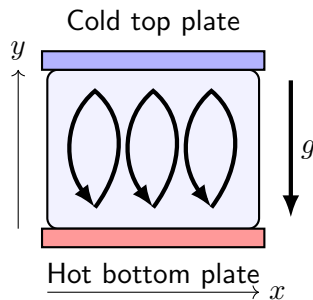
Rayleigh–Bénard Convection

Physical Setting

We consider a horizontal layer of fluid confined between two plates: the lower plate is kept hot and the upper plate cold. Gravity acts downward, so warmer (lighter) fluid tends to rise while colder (denser) fluid tends to sink. Initially heat is transferred purely by conduction, but beyond a certain threshold the fluid becomes unstable and begins to move, forming circulating convection cells.

Where This Occurs in Nature and Technology

Rayleigh–Bénard convection is a fundamental mechanism in many systems: atmospheric and oceanic circulation, mantle convection under the Earth's crust, heat transfer in engineering devices, cooling systems, and even thin liquid films in industrial processes.



Notation and Physical Parameters

Fields

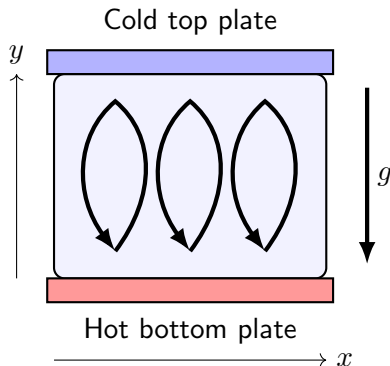
- ▶ $u = (u_x, u_y)$ – velocity field
- ▶ $T(x, y, t)$ – temperature field

Material Parameters

- ▶ ν – kinematic viscosity
- ▶ κ – thermal diffusivity
- ▶ α – thermal expansion coefficient
- ▶ g – gravitational acceleration

Dimensionless Numbers

- ▶ Rayleigh number: $Ra = \frac{g\alpha \Delta T L_y^3}{\nu\kappa}$
- ▶ Prandtl number: $Pr = \frac{\nu}{\kappa}$



Governing Equations: Boussinesq Approximation

Assumptions (Boussinesq approximation)

- 1) **Momentum balance:** density is nearly constant ($\rho \approx \rho_0$) except in the buoyancy term $\rho = \rho_0(1 - \alpha(T - T_0))$; viscosity provides momentum diffusion via $\nu \nabla^2 u$.
- 2) **Energy balance:** temperature evolves through advection by the flow and thermal diffusion with diffusivity κ .
- 3) **Mass conservation:** the fluid is incompressible, so $\nabla \cdot u = 0$.

Boussinesq Equations (derived from conservation laws)

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\nabla p + \nu \nabla^2 u + g\alpha (T - T_0) \mathbf{e}_y.$$

$$\frac{\partial T}{\partial t} + (u \cdot \nabla)T = \kappa \nabla^2 T.$$

$$\nabla \cdot u = 0.$$

Dimensionless Boussinesq Equations

Dimensionless temperature

Scale temperature by the imposed temperature difference

$$\Delta T = T_{\text{bottom}} - T_{\text{top}},$$

and define the dimensionless temperature

$$\theta(x, y, t) = \frac{T(x, y, t) - T_{\text{top}}}{\Delta T}.$$

Final non-dimensional system

$$\partial_t \theta + (u \cdot \nabla) \theta = \nabla^2 \theta,$$

$$\partial_t u + (u \cdot \nabla) u = -\nabla p + \text{Pr} \nabla^2 u + \text{Ra Pr} \theta \mathbf{e}_y,$$

$$\nabla \cdot u = 0.$$

Boundary conditions

1) $\theta(y = 0) = 1, \theta(y = 1) = 0$

2) No-slip velocity at the plates:
 $u(y = 0) = u(y = 1) = 0$

3) Horizontal direction is periodic

From Conduction to Complex Convection

Low Rayleigh number: no motion

When the temperature difference is small (low Ra), the fluid stays still. Heat is transported only by conduction, and the temperature profile is linear.

Moderate Rayleigh number: convection rolls

Increasing Ra makes the conductive state unstable. The flow organizes itself into large, steady convection rolls that circulate hot fluid upward and cold fluid downward.

High Rayleigh number: time dependence and chaos

For sufficiently large Ra , the rolls begin to drift, merge, break up, and form increasingly irregular patterns. The system becomes time-dependent and eventually chaotic.

Why this matters for ML

These complex, multiscale patterns are expensive to simulate numerically and difficult to predict analytically — making Rayleigh–Bénard convection an ideal testbed for data-driven surrogate models.

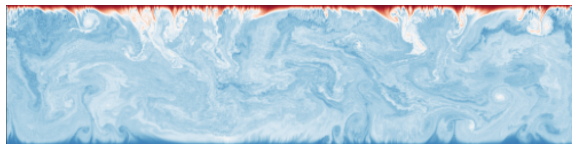
Dataset: Polymathic AI Rayleigh–Bénard Convection

What the dataset contains

Simulations generated by a high-resolution spectral solver. Each sample contains full fluid fields: $b(x, y, t)$, $u_x(x, y, t)$, $u_y(x, y, t)$, $p(x, y, t)$. Here b is the buoyancy (proportional to temperature deviation).

Resolution and structure

- ▶ Spatial grid: 512×128
- ▶ Temporal length: 200 timesteps
- ▶ Time step: $\Delta t = 0.25$ (non-dimensional)
- ▶ Vertical direction: Chebyshev nodes
- ▶ Horizontal direction: uniform grid, periodic



Parameter coverage

The dataset spans many combinations of Rayleigh and Prandtl numbers:

$$\text{Ra} \in \{10^6, 10^7, 10^8, 10^9, 10^{10}\},$$

$$\text{Pr} \in \{0.1, 0.2, 0.5, 1, 2, 5, 10\}.$$

For each (Ra, Pr) there are multiple runs with different initial conditions.

Target Quantity: Local Buoyancy Flux

Why we need a derived field

The dataset provides several raw fields (b, u_x, u_y, p) . Predicting all of them directly is unnecessary and expensive. Instead, we focus on a physically meaningful diagnostic that captures convection strength and heat transport.

Temperature deviation

Let

$$\bar{b}(y, t) = \langle b(x, y, t) \rangle_x, \quad \theta_b(x, y, t) = b(x, y, t) - \bar{b}(y, t),$$

where θ_b measures fluctuations relative to the horizontal mean (standard in RBC analysis).

Local buoyancy flux

The quantity of interest is

$$j_b(x, y, t) = u_y(x, y, t) \theta_b(x, y, t),$$

an analogue of the convective heat flux used widely in turbulence studies.

Objective: Predict Buoyancy Flux Field

Learning objective

Given a sequence of past fields

$$j_b(\cdot, \cdot, t_0), j_b(\cdot, \cdot, t_1), \dots, j_b(\cdot, \cdot, t_T),$$

the goal is to predict future fields

$$j_b(\cdot, \cdot, t_{T+1}), \dots, j_b(\cdot, \cdot, t_{T+k}).$$

High-level strategy

- ▶ Compress each 2D field into a low-dimensional representation.
- ▶ Learn temporal evolution of this representation.
- ▶ Reconstruct future j_b fields from predicted latent states.

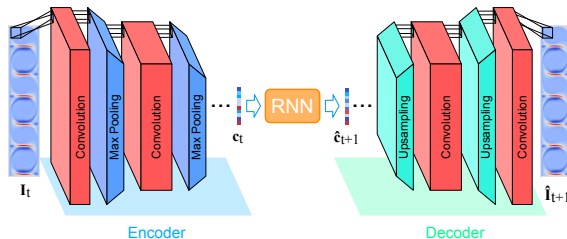
This reduces the high-dimensional, chaotic PDE dynamics into a tractable sequence modeling problem.

Model Architecture¹

Model the temporal evolution of the buoyancy-flux field $j_b(x, y, t)$ by combining:

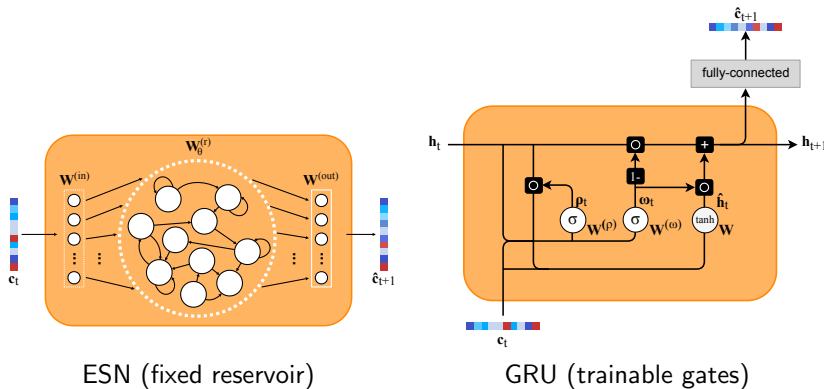
- ▶ A **convolutional encoder** to compress each field into a low-dimensional latent vector c_t ,
- ▶ An **RNN** that predicts the next latent state $\hat{c}_{t+1}$ from c_t ,
- ▶ A **convolutional decoder** that reconstructs the future field $\hat{j}_b(x, y, t + 1)$ from the predicted latent code.

$$j_b(\cdot, \cdot, t) \xrightarrow{\text{Encoder}} c_t \xrightarrow{\text{RNN}} \hat{c}_{t+1} \xrightarrow{\text{Decoder}} \hat{j}_b(\cdot, \cdot, t + 1).$$



¹Sandeep Pandey et al. Direct data-driven forecast of local turbulent heat flux in Rayleigh–Bénard convection. *Physics of Fluids* 34.4 (2022).

Recurrent Module: ESN vs. GRU



ESN (fixed reservoir)

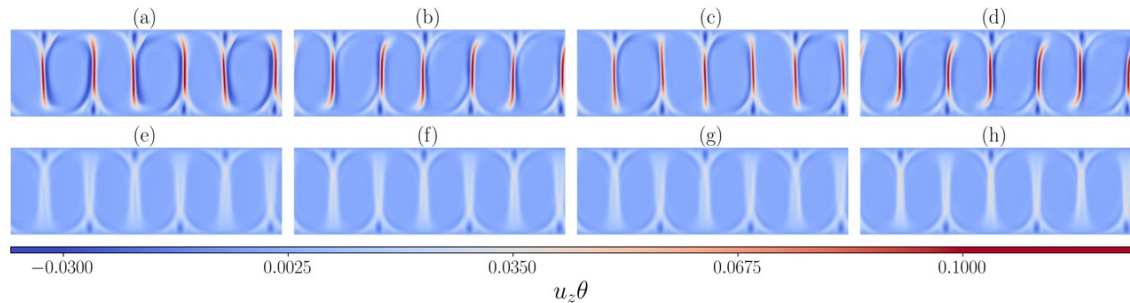
GRU (trainable gates)

ESN uses a fixed random reservoir updated by

$$r_t = (1 - \alpha) r_{t-1} + \alpha \tanh\left(W^{(in)} c_t + W_{\eta}^{(r)} r_{t-1}\right),$$

training only the output weights, while GRU learns all recurrent parameters end-to-end.

Results



(a)–(d) Instantaneous convective turbulent heat flux snapshots at a given time.

(e)–(h) Mean convective turbulent heat flux field averaged over 900 snapshots.

(a, e) DNS, i.e., ground truth.

(b, f) CAE, i.e., ground truth for the ESN and GRU.

(c, g) Prediction from the ESN with subsequent decoding.

(d, h) Prediction from the GRU with subsequent decoding.